

PRANJAL SWARUP BHATNAGAR

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EXPERIENCE

Associate, Norman Lab

Next Tech Lab (QS Reimagine Education Award winner)

Apr 2026 - Present
SRMIST, Kattankulathur

- Selected by interview into Norman Lab at Next Tech Lab, the first student-led organisation from India to win the QS Reimagine Education Award.
- Built StupidHacks, the portal for the lab's internal hackathon, with a small team. Owned the data layer and its rule tests, the design system on the lab's brand, and the move to Supabase with one-time-code roster sign-in. Live at stupidhack.nexttechlab.in.

PROJECTS

[ExamDB](#) | Python, FastAPI, PostgreSQL, Next.js, Docker

- Built and shipped a searchable database of **194** Indian competitive exams, consolidating dates scattered across dozens of government portals into one source that cites the official page behind every date.
- Designed a scrape/normalise/dedup/reduce pipeline that preserves conflicting sources and flags the disagreement instead of silently picking a winner, so wrong data stays traceable to the scraper that produced it.
- Caught a scraper that passed its own tests while emitting **46** rows of a historical archive stamped with the current year, spotted from an impossible 29 FEB entry. Added source-diff validation so a row count is never mistaken for correctness.
- Built OAuth sign-in storing only SHA-256 digests of opaque session tokens, plus daily scrapers on GitHub Actions and a Next.js frontend. Solo project, live at examdb.org.

[TYPE](#) | Rust, ratatui, LSP, DAP, tree-sitter

- Building a full IDE that runs in the terminal, non-modal with full mouse parity, aimed at the gap between bare terminal editors and heavyweight GUI ones. Published on crates.io as typ-editor, currently **v0.2.10**.
- Wrote LSP, DAP and tree-sitter clients so new language support arrives by protocol instead of per-language code. Holds a **100ms** cold-start and **16ms** p99 keystroke-to-glyph budget.
- Added an opt-in vim layer that maps modes and operators onto the same named actions the non-modal path already calls, so it owns no editing primitives and the core never depends on it.
- Made themes a TOML artifact rather than a compile-time constant, detecting terminal colour depth at startup and degrading the palette to 256 colours where required.

[OutN](#) | Python, discord.py, OpenCV

- Self-hostable FOSS Pokemon recognition and assistant bot for Discord, identifying spawns from images and solving hints automatically; config-driven so users deploy it without writing code.
- Maintained across **2+ years** and **9 major versions** through repeated breaking changes to the upstream Discord and Poketwo APIs, keeping existing self-hosted deployments working without manual migration.

OPEN SOURCE

[onthespot](#) | Python, "1.7k stars"

- Fixed broken Bandcamp search in a widely-used multi-service music downloader (**1.7k stars**) by porting it onto the provider's public search API, plus an AppImage runtime-resolution fix. Three PRs merged upstream.

EDUCATION

[SRM Institute of Science and Technology](#)

M.Tech (Integrated) Computer Science, Cybersecurity & Digital Forensics

2025 - 2030

Delhi Public School

Class XII

2011 - 2024

SKILLS

Languages	Python, Rust, TypeScript, JavaScript, Go, SQL
Backend	FastAPI, SQLAlchemy, PostgreSQL, Hono, Express
Frontend	Next.js, React, Svelte, Astro, TailwindCSS
Infrastructure	Docker, GitHub Actions, Vercel, Supabase, Redis, Git